

Taj Dyson

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Experimental physicist studying axion dark matter using cryogenic quantum devices.

EDUCATION

Stanford University <i>PhD, Physics (GPA 3.90)</i>	Sep. 2021 – Jun. 2027
McGill University <i>Bachelor of Science, Honours Physics (GPA 3.76)</i>	Aug. 2018 – May 2021
Dawson College <i>DCS, First Choice Sciences, Honours List</i>	Aug. 2016 – May 2018

SELECTED AWARDS

NSERC Postgraduate Scholarships – Doctoral	2023 – 2026
Robert H. Siemann Fellowship	2022 – 2025
NSERC Undergraduate Summer Research Award & FRQNT Supplement	2020
BLUE Fellowship at McGill’s Building 21	2020
McGill Physics Hackathon Winner – Arts & Science	2019

SELECTED PUBLICATIONS

(Click entry to view, see below for full list)

- H. Magoon, T. Aralis, **T. Dyson**, *et. al.* “A First Demonstration of the SQUAT Detector Architecture: Direct Measurement of Resonator-Free Charge-Sensitive Transmons,” *In submission for Physical Review Applied* (2026).
- G. Bratrud, S. Lewis, K. Anyang, **T. Dyson**, *et. al.* “Measurement of Correlated Charge Noise in Superconducting Qubits at an Underground Facility,” *Nature Communications* (2025)
- **T. Dyson**, C. Bartram, A. Davidson, *et al.*, “High-volume Tunable Resonator for Axion Searches Above 7 GHz,” *Phys. Rev. Applied* (2024).
- **T. Dyson**, H. C. Chiang, E. Egan, *et al.*, “Radio-Frequency Interference at the McGill Arctic Research Station,” *Journal of Astronomical Instrumentation* (2021).

SELECTED PRESENTATIONS AND OUTREACH

(Click entry to view, see below for full list)

Design and Characterization of SQUAT Detectors <i>Talk, APS Global Physics Summit</i>	Mar. 2026
High-frequency Haloscopes for the ADMX-VERA Program <i>Talk, APS Global Physics Summit</i>	Mar. 2025
Resonating with Dark Matter with ADMX-VERA <i>Article, Stanford KIPAC Research Highlights</i>	Dec. 2024
Developing ADMX-VERA, an Axion Haloscope Above 7 GHz <i>Talk, Fermilab Superconducting Quantum Materials and Systems Seminar</i>	Mar. 2024
Radio Frequency Interference at the McGill Arctic Research Station <i>Talk, Soup and Science Public Talks, McGill University</i>	Sep. 2020
Interviewed in “ALBATROS radio astronomy Product Showcase” <i>Article, The MagPi Magazine</i>	Sep. 2019

Graduate Research Assistant – Axion Haloscope

Sep. 2021 – Present

Stanford University, supervised by Prof. Chao-Lin Kuo

- Characterized the resonances of a novel prototype haloscope resonator for axion and dark photon dark matter detection.
- Developed an algorithm to automatically align the haloscope across five degrees of freedom based on measurements possible at cryogenic temperatures.
- Performed precision metrology on haloscope parts in a custom-built LN₂ bath.
- Designed and commissioned a cryogenic dark photon search featuring a 7-kg moving part with six degrees of freedom, including a vacuum system extension, thermal sinking, and radiation shielding.
- Performed a tabletop dark photon search using custom control software and a SQL database, with analysis using Python. The forthcoming results will set leading constraints on dark photons with mass 21–29 μeV .
- Mentored four undergraduates, with weekly meetings to set goals and assess challenges as well as collaborative work in the laboratory.

Graduate Research Assistant – Qubit-Based Sensors

Mar. 2022 – Present

SLAC and Stanford University, supervised by Dr. Noah Kurinsky

- Took continuous-wave and pulsed measurements to characterize superconducting quasiparticle-amplifying transmons (SQUATs), a novel low-energy rare event detector.
- Simulated SQUAT data to optimize the choice of integration time for given device properties.
- Designed and integrated a hardware system for actuating cryogenic switches for radio frequency cables, and wrote a Python library to interface with it. These switches enable new types of cryogenic measurement and allow multiple devices to be tested without cycling the cryostat.
- Characterized a superconducting traveling wave parametric amplifier (TWPA), finding optimal operating parameters and its noise temperature.
- Learned firsthand to operate a dilution refrigerator.

Graduate Research Assistant – Axion Dark Matter Experiment

Feb. 2023 – Present

Stanford University, supervised by Prof. Chao-Lin Kuo for the ADMX Collaboration

- Sped up analysis of data taken in high-resolution mode by employing pre-compiled size- and type-aware functions rather than interpreted Python.
- Coordinated across the collaboration to gather past simulations and measurements done for the proposed ADMX Extended Frequency Range experiment, for publication of a detailed design study.
- Operated the ADMX main experiment during science data taking, including Josephson parametric amplifier pump optimization and rod actuation for re-scanning.

Graduate Research Assistant – Cosmic Microwave Background Receivers

May 2022 – Jul. 2023

Stanford University, supervised by Prof. Chao-Lin Kuo

- Took cryogenic measurements of the noise performance of a novel detector type, thermal kinetic inductance detectors (TKIDs).
- Contributed to design the cryogenic radio-frequency readout chain for a receiver in the BICEP array.

Undergraduate Research Assistant – Radio Cosmology Field Work

Sep. 2018 – Jul. 2021

McGill University, supervised by Prof. Cynthia Chiang

- Developed and deployed solar and wind power solutions for radio interferometer stations in remote locations such as Uapishka Station and the McGill Arctic Research Station.
- Designed & built electronic devices and wrote C++ Arduino firmware for power control & logging.
- Flagged radio-frequency interference in radio data using custom Python scripts.

TEACHING EXPERIENCE

Volunteer Instructor

Jun. 2023 – Jun. 2024

Virtù Institute at Stanford University

- Designed, procured, and hand-assembled a magnetic track to levitate a superconducting puck as a physics demonstration.
- Designed and delivered a lesson on magnetism and superconductivity for kids aged 12 to 18 with demonstrations and interactive activities.
- Contributed to overall day camp lesson plans and volunteer co-ordination.

Teaching Assistant – Intro. Heat & Light

Sep. 2023 – Dec. 2023

Stanford University

- Volunteered to refurbish, document, and present in-class physics demonstrations of thermodynamic and optical phenomena on a twice-weekly basis.
- Prepared and delivered lessons in “discussion sections,” small-class active learning sessions with worksheets.
- Graded assignments and exams.

Teaching Assistant – Intro. Electricity and Magnetism

Jan. 2023 – Mar. 2023

Stanford University

- Taught in a teaching laboratory, prepared and led study sessions and office hours, graded assignments and exams.

Teaching Assistant – Intro. Mechanics

Jan. 2022 – Mar. 2022

Stanford University

- Taught in an active learning classroom, prepared and led study sessions and office hours, graded assignments and exams.

TEAM Undergraduate TA

Jan. 2021 – May 2021

McGill University

Advanced Undergraduate – Data Science and Observational Astrophysics

- Mentored students through coding labs in an online active learning environment.

Instructor

Feb. 2017 – Sep. 2017

Kids Code Jeunesse

- Taught introductory coding to kids of all ages in several workshops and events, including Scratch, HTML, and Python.

ACADEMIC SERVICE

Camping Czar

Jun. 2022 – May 2023

Stanford University Graduate Students of Applied Physics and Physics

- Managed and delegated campsite booking, meal procurement, and transport for two cross-departmental weekend camping trips. Typical attendance was 50 people.

Vice President

Sep. 2023 – Nov. 2024

Stanford University Graduate Students of Applied Physics and Physics

- Held meetings to co-ordinate the physics graduate student society’s volunteer officers, covering the Stanford physics open house, orientation week, and other student events.

TECHNICAL PROFICIENCIES

Design & Simulation: SOLIDWORKS, KiCAD, COMSOL, HFSS

Manufacturing: Manual mill, manual lathe, waterjet cutter, laser cutter/engraver, soldering iron, 3D printer, coordinate measuring machine (CMM) for metrology, wet bench for copper passivation

Laboratory: Dilution refrigerator operation, vector network analyzer (VNA) measurement, spectrum analyzer measurement, high-vacuum system assembly and helium leak checking

Software tools: SQL databases, Pyvisa experiment control, local networking

Languages: Python, bash, Perl, C, C++, Lua, Julia

GRADUATE-LEVEL COURSES

Quantum Field Theory I	<i>Stanford University</i>
Cosmology	<i>Stanford University</i>
Physics of Energy	<i>Stanford University</i>
Quantum Theory	<i>McGill University</i>
Electromagnetic Theory	<i>McGill University</i>
General Relativity	<i>McGill University</i>
Particle Physics	<i>McGill University</i>
Advanced Statistical Mechanics	<i>McGill University</i>
Biophysics	<i>McGill University</i>

COMPLETE PUBLICATIONS

- H. Magoon, *et al.* (incl. **T. Dyson**), “A First Demonstration of the SQUAT Detector Architecture: Direct Measurement of Resonator-Free Charge-Sensitive Transmons,” *arXiv:2601.16261* (2026).
- C. Goodman, *et al.* (incl. **T. Dyson**), “ADMX Axion Dark Matter Bounds around $3.3 \mu\text{eV}$ with Dine-Fischler-Srednicki-Zhitnitsky Discovery Ability,” *Phys. Rev. Lett.* (2025).
- G. Carosi, *et al.* (incl. **T. Dyson**), “Search for Axion Dark Matter from 1.1 to 1.3 GHz with ADMX,” *Phys. Rev. Lett.* (2025).
- G. Bratrud, *et al.* (incl. **T. Dyson**), “Measurement of Correlated Charge Noise in Superconducting Qubits at an Underground Facility,” *Nature Communications* (2025).
- A. T. Hipp, *et al.* (incl. **T. Dyson**), “Search for Non-virialized Axions with $3.3\text{--}4.2 \mu\text{eV}$ Mass at Selected Resolving Powers,” *Phys. Rev. D* (2025).
- C. L. Kuo, *et al.* (incl. **T. Dyson**), “Maximizing Quantum Enhancement in Axion Dark Matter Experiments,” *Phys. Rev. D* (2025).
- M. Guzzetti, *et al.* (incl. **T. Dyson**), “Improved Receiver Noise Calibration for ADMX Axion Search: 4.54 to $5.41 \mu\text{eV}$,” *Phys. Rev. D* (2025).
- T. Dyson**, *et al.*, “High-volume Tunable Resonator for Axion Searches Above 7 GHz,” *Phys. Rev. Applied* (2024).
- T. Dyson**, *et al.*, “Radio-Frequency Interference at the McGill Arctic Research Station,” *Journal of Astronomical Instrumentation* (2021).